Abstract

AttendX -A Smart Face Recognition and Location-Based Attendance System is a modern, mobile-based automated attendance management system developed to address the limitations of traditional manual attendance methods used in educational institutions. Conventional attendance systems are time-consuming, prone to errors, and susceptible to proxy attendance. AttendX aims to overcome these challenges by integrating machine learning, facial recognition, time validation, and location-based verification into a single secure platform.

The proposed system uses a mobile application through which students mark their attendance by capturing a live facial image during predefined time intervals corresponding to college sessions. The system employs machine learning-based face detection and face recognition models to accurately identify students. During initial registration, multiple facial samples of each student are collected and converted into numerical face embeddings, which are securely stored in the database. At attendance time, the captured face is matched against the stored embeddings to verify the student's identity.

To further enhance reliability and prevent misuse, AttendX incorporates GPS-based geofencing to ensure that attendance is marked only when the student is physically present within the college premises. Additionally, time window validation ensures that attendance can be recorded only during authorized class hours. A liveness detection mechanism, such as eye blink and smile detection, is implemented to prevent attendance marking using photographs or recorded videos.

Once all conditions—face match, valid time, valid location, and liveness—are satisfied, attendance is automatically recorded along with the timestamp and location details. The system also provides a teacher dashboard that generates daily and weekly attendance reports, attendance percentages, and summaries, thereby reducing faculty workload and improving accuracy.

AttendX is a cost-effective solution as it utilizes existing smartphones and open-source technologies, eliminating the need for dedicated biometric hardware. The system is scalable, secure, and suitable for real-world academic environments. By automating attendance management, AttendX enhances efficiency, minimizes human intervention, and provides a reliable digital solution for modern educational institutions.